Federico Battista

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Contact Information

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Viterbi School of Engineering
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Research Interests

Discrete optimization, spanning theory, algorithms, and computation in the development of solution methods for complex mixed-integer linear problems, including those arising in **bilevel optimization** and **combinatorial optimization**, and on the development of open-source software for use in real-world decision problems.

Academic Experience

Lehigh University,
Dept. of Industrial and Systems Engineering, Bethlehem, PA, USA

Adjunct Professor — *January 2027 – May 2027*

University of Southern California,
Daniel J. Epstein Dept. of Industrial and Systems Engineering, Los Angeles, CA, USA

Postdoctoral Researcher under Karmel S. Shehadeh — *February 2026 – January 2027*

Lehigh University,
Dept. of Industrial and Systems Engineering, Bethlehem, PA, USA

Postdoctoral Researcher under Ted Ralphs — *February 2023 – January 2026*

Education

Università di Roma Sapienza
Dept. of Computer, Control, and Management Engineering, Rome, Italy

Ph.D. in Operations Research — *November 2019 – January 2023*
Thesis: On Semidefinite Lift-and-Project Relaxations of Combinatorial Optimization Problems
Advisors: Marianna De Santis, Fabrizio Rossi, Stefano Smriglio

Alpen-Adria-Universität
Institut für Mathematik, Klagenfurt, Austria

Visiting Ph.D. student under Angelika Wiegele — *February 2022 – July 2022*

Università degli Studi dell'Aquila
Dept. of Information Engineering, Computer Science and Mathematics, L'Aquila, Italy

Master's degree in Computer Science, *Magna Cum Laude* — *July 2019*
Bachelor's degree in Computer Science, *Magna Cum Laude* — *October 2017*

Refereed Publications

[1] F. **Battista**, F. Rossi, and S. Smriglio. *Application of the Lovász-Schrijver Operator to Compact Stable Set Integer Programs*. In: *Mathematical Programming Computation* (2025). DOI: 10.1007/s12532-025-00298-8.

[2] F. **Battista** and M. De Santis. *Dealing with Inequality Constraints in Large-Scale Semidefinite Relaxations for Graph Coloring and Maximum Clique Problems*. In: *4OR. A Quarterly Journal of Operations Research* (2024), pp. 1–31. DOI: 10.1007/s10288-024-00569-5.

Working Papers

[3] F. **Battista**, F. Rossi, and S. Smriglio. *Linear versus Semidefinite Lovász–Schrijver Lifting Operators of Stable Set Integer Programs*. Working paper. 2026.

[4] K. S. Shehadeh, M. Y. Tsang, and F. **Battista**. *Contextual Residual-Based Distributionally Robust Optimization under Predictive Model Uncertainty*. Working paper. Dept. of Industrial and Systems Engineering, University of Southern California, 2026.

[5] F. **Battista** and T. K. Ralphs. *Improving Directions in Mixed Integer Bilevel Linear Optimization*. arXiv preprint. COR@L Laboratory, Lehigh University, 2025. URL: https://doi.org/10.48550/arXiv.2511.03566. *Under review at Mathematical Programming Computation*.

[6] S. Fallah, F. **Battista**, and T. K. Ralphs. *A Branch-and-Bound Algorithm for Multiobjective Mixed Integer Linear Optimization*. Working paper. COR@L Laboratory, Lehigh University, 2025.

Software

[7] F. **Battista**, F. Rossi, and S. Smriglio. *SDP_lift_and_project* (2025). URL: https://github.com/febattista/SDP_lift_and_project. *Maintainer*.

[8] F. **Battista** and M. De Santis. *ADAL-Ineq* (2024). URL: https://github.com/febattista/ADAL-ineq. *Maintainer*.

[9] S. DeNegre, T. Ralphs, and S. Tahernejad. *MibS Version 1.2* (2024). URL: https://github.com/coin-or/MibS. *Contributor*.

[10] T. Ralphs, M. Güzelsoy, and A. Mahajan. *SYMPHONY Version 5.7* (2023). URL: https://github.com/coin-or/SYMPHONY/. *Contributor*.

Students

Doctoral Students Mentorship

- Shannon Kelley, Lehigh University, ISE, Advisor: Ted Ralphs — *2023 – 2026*
- Samira Fallah, Lehigh University, ISE, Advisor: Ted Ralphs — *2023 – 2026*
- Hesam Shaelaie, Lehigh University, ISE, Advisor: Ted Ralphs — *2023 – 2026*
- Suresh Bolusani, Lehigh University, ISE, Advisor: Ted Ralphs — *2023 – 2026*
- Yu Xie, Lehigh University, ISE, Advisor: Ted Ralphs — *2023 – 2026*

Teaching Experience

Instructor

- ISE 308/408 Simulation, undergraduate and graduate course, Lehigh University — *2027*

Guest Lecturer

- ISE 418 Discrete Optimization, Ph.D. course, Lehigh University (4 lectures) — *2024*

Teaching Assistant

- Operations Research Laboratory, B.Sc. course, Università Roma Sapienza — *2020*
- Advanced Calculus, B.Sc. course, Università Roma Sapienza — *2020*

Talks

11. F. **Battista** and T. K. Ralphs. *The Role Of Improving Directions in Mixed-Integer Bilevel Linear Optimization*. In: *Bilevel Optimization Society Webinar Series* (2026). **Invited Talk**.
12. F. **Battista**, T. K. Ralphs, and S. Fallah. *The Restricted Value Function of MILPs and Its Construction with SYMPHONY*. In: *18th INFORMS Computing Society (ICS) Conference* (2025).
13. F. **Battista** and T. K. Ralphs. *A Branch-and-Cut Algorithm for Mixed-Integer Bilevel Linear Optimization Based on Improving Directions*. In: *International Symposium on Mathematical Programming* (2024).
14. F. **Battista** and T. K. Ralphs. *Exploiting Dual Functions in Mixed Integer Bilevel Linear Programs*. In: *INFORMS Annual Meeting* (2023).
15. F. **Battista**, F. Rossi, and S. Smriglio. *Application of the Lovász-Schrijver Operator to Representative Formilation for Graph Coloring Problem*. In: *International Symposium on Combinatorial Optimization* (2022).
16. F. **Battista** et al. *On Semidefinite Lift-and-Project Relaxations for Combinatorial Optimization Problems*. In: *University of Pavia* (2022).
17. F. **Battista** and M. De Santis. *Dealing with Inequalities in Large Scale Semidefinite Programs*. In: *International Conference on Optimization and Decision Science* (2021).

Service

Editorial Board (Associate Editor)

- Operations Research (Area: Data, Software, and Computation) *2025 – Present*